

infrared.city-1

Tree Detection



**WIRTSCHAFTS
UNIVERSITÄT
WIEN VIENNA
UNIVERSITY OF
ECONOMICS
AND BUSINESS**

Moritz Hörmansdorfer
Sara Klasová
Dominik Oberhumer
Nina Salnikow

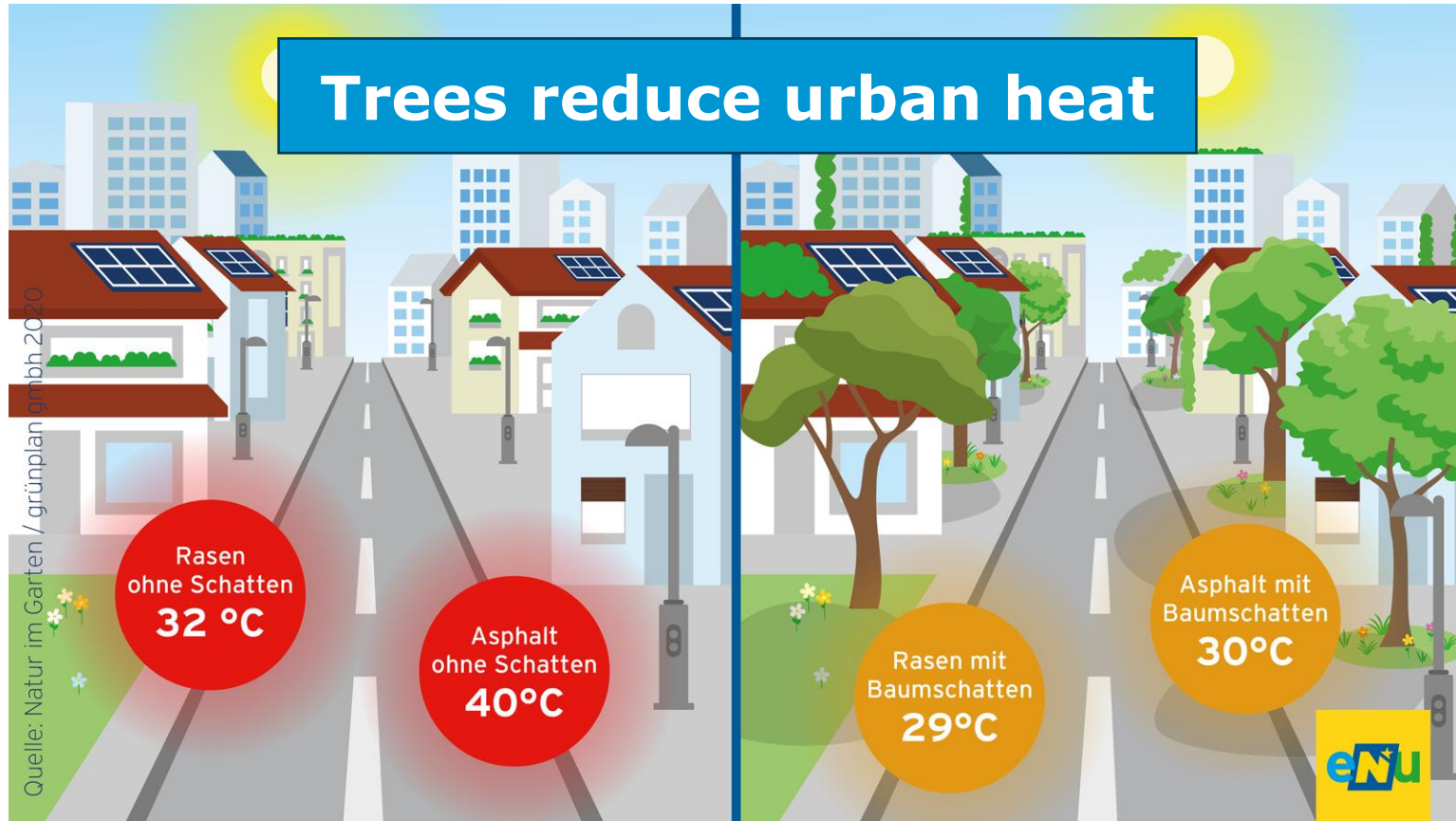
Agenda

- Introduction
- Project Outline
- Data
- Exploratory Data Analysis
- Machine Learning Pipeline
- Current Problems
- Follow Up

Introduction



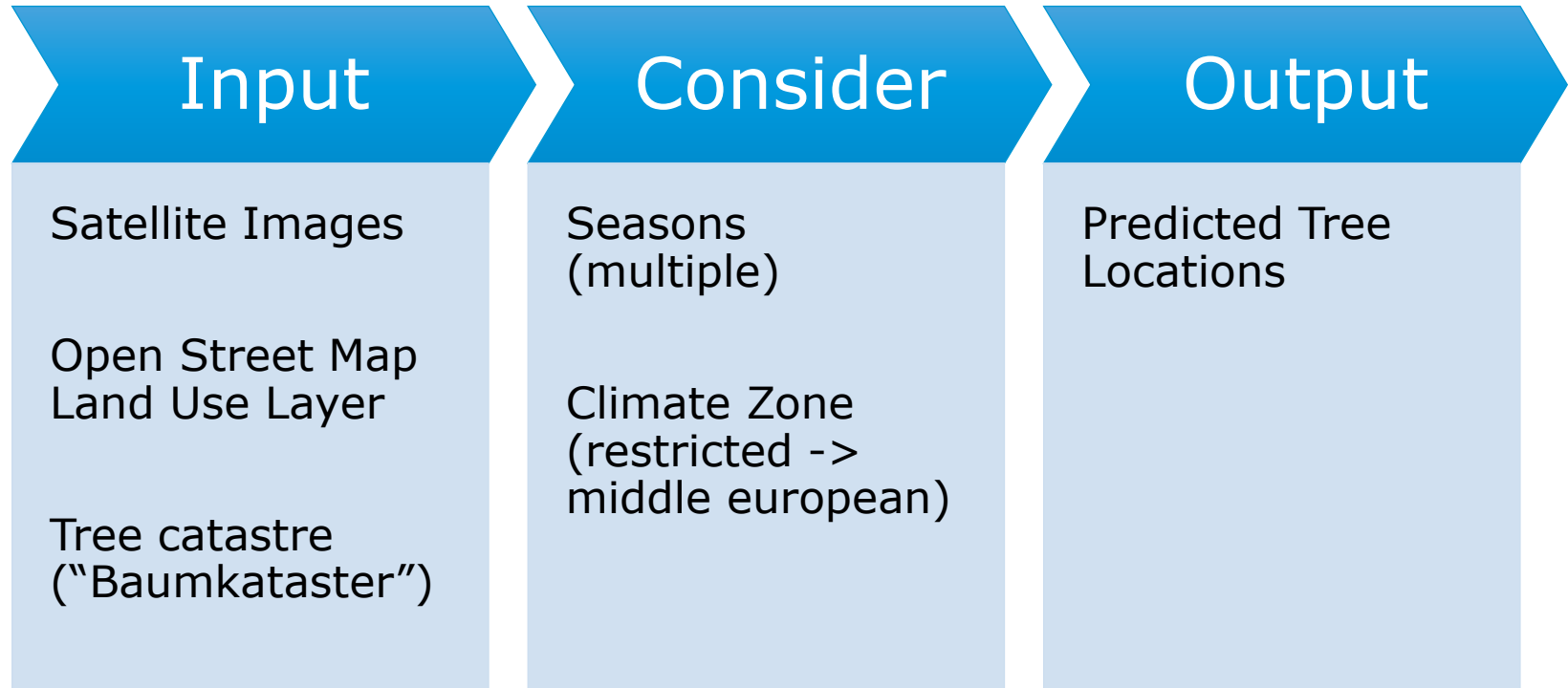
Introduction - Problem



- Infrared.city
 - Climate simulation platform
 - AI-powered
 - Simulates: thermal stress, wind comfort, etc.
 - Used by: architects, city government, etc.
- Previous projects
 - Detected Flood Risk
 - Detected Greenspace



Introduction – Project Outline



Data



- Tree cadastre data
(Vienna, Paris, Barcelona, Hamburg)
- Sentinel-2 data
- OpenStreetMap data

- ~6GB dataset collected

```
jovyan@jupyter-h12109432:~/ideas-dsl  
ab-group1-shared$ du -sh "raw data"  
5.6G    raw data
```

- Cleaned & standardized (e.g. rename columns)

Approach & Challenge

- **Previous project:** Manual image selection
- **Initial approach:** Copernicus API

→ **Switched to Google Earth Engine**

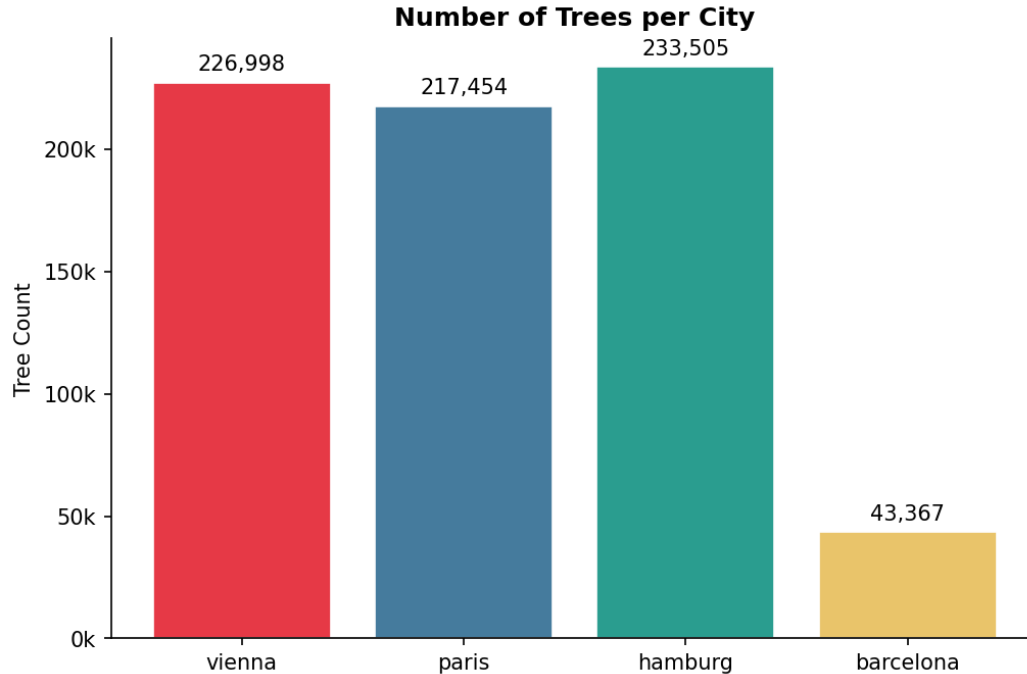
- **Challenge:** Cloud coverage filtering tuning

Exploratory Data Analysis



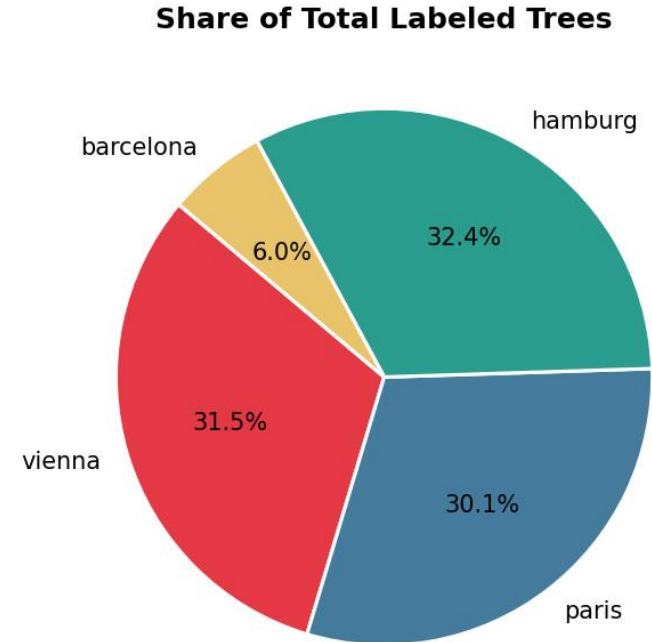
EDA - Baumkataster

Baumkataster - City Overview



Training cities are balanced

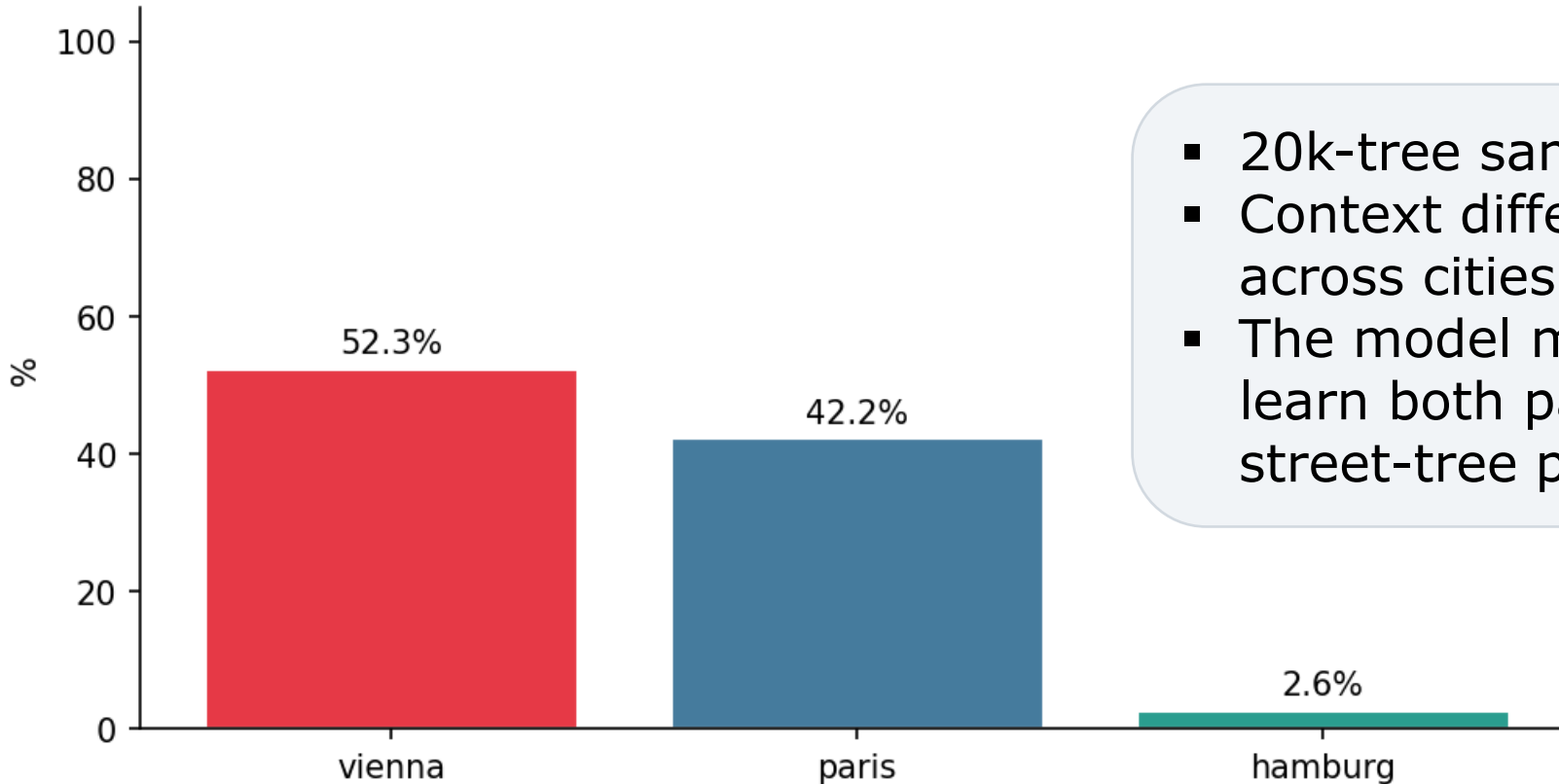
Vienna / Paris / Hamburg are large enough for joint training and comparison



Barcelona is a hold-out case

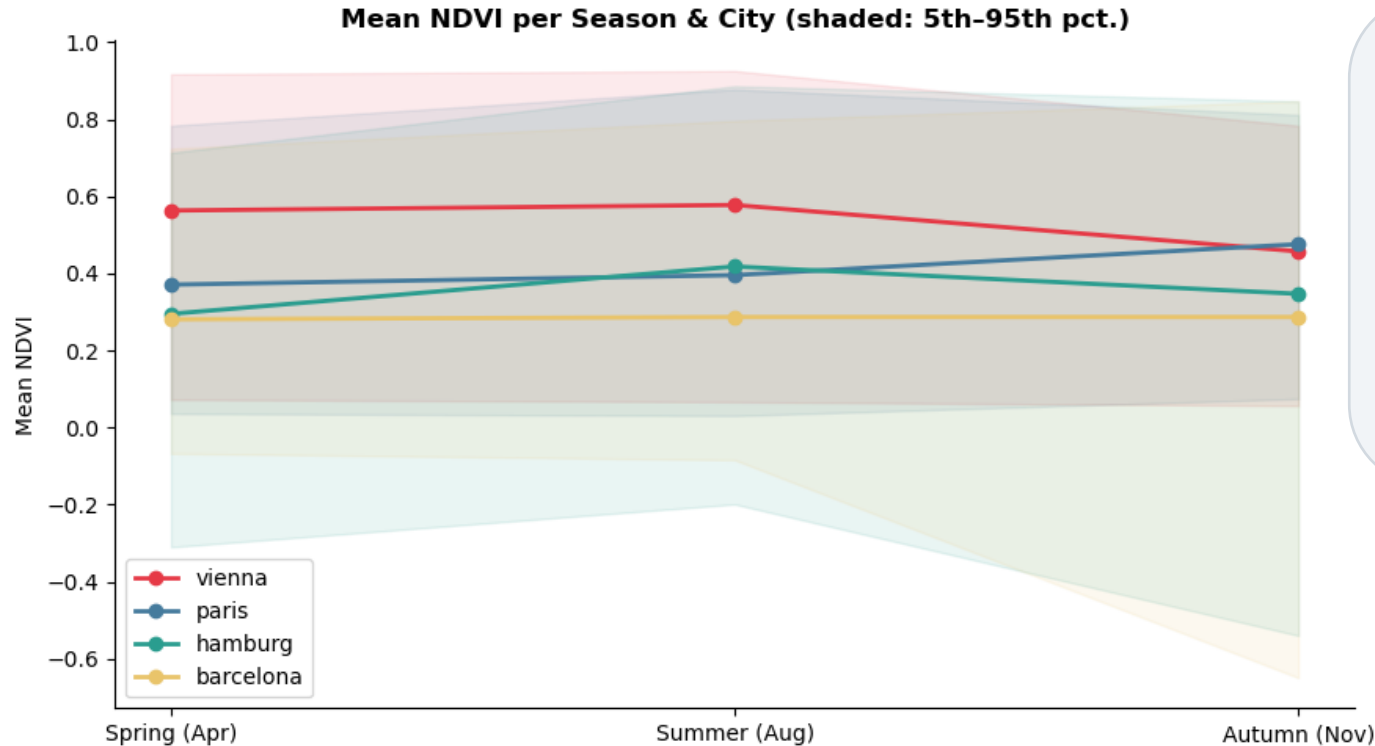
Only 43k labels, no geometry / size attributes — useful as a generalization test

% of Trees Located Inside Green Landuse Zones (park, forest, grass, meadow, scrub)



- 20k-tree sample
- Context differs across cities
- The model must learn both park and street-tree patterns

EDA – Sentinel



Seasonal peak differs by city

- Vienna/Hamburg peak in summer
- Paris peaks in autumn
- Barcelona nearly flat

Multi-season differences add signal, but not with one shared pattern across all cities

EDA - Data quality checks

band		EVI	NDVI
city	season		
barcelona	apr	0.9000	0.2807
	aug	0.8557	0.2867
	nov	0.9689	0.2867
hamburg	apr	0.8083	0.2949
	aug	1.2413	0.4179
	nov	0.7093	0.3469
paris	apr	1.3363	0.3708
	aug	1.3314	0.3955
	nov	1.4568	0.4757
vienna	apr	1.6667	0.5630
	aug	1.7176	0.5774
	nov	0.9960	0.4571

1. EVI / SAVI are mis-scaled

- Observed values exceed 1.0, which is impossible for these indices.
- Points to the formula being applied to raw integer bands (0–10,000) rather than reflectance

**Raw
bands
0–
10,000**



**Formula
applied
directly**



**Inflated
EVI /
SAVI
values**

**Fix: divide bands by 10,000 and recompute
EVI/SAVI from the existing TIF stacks
NDVI is safe as-is**

EDA - Data quality checks

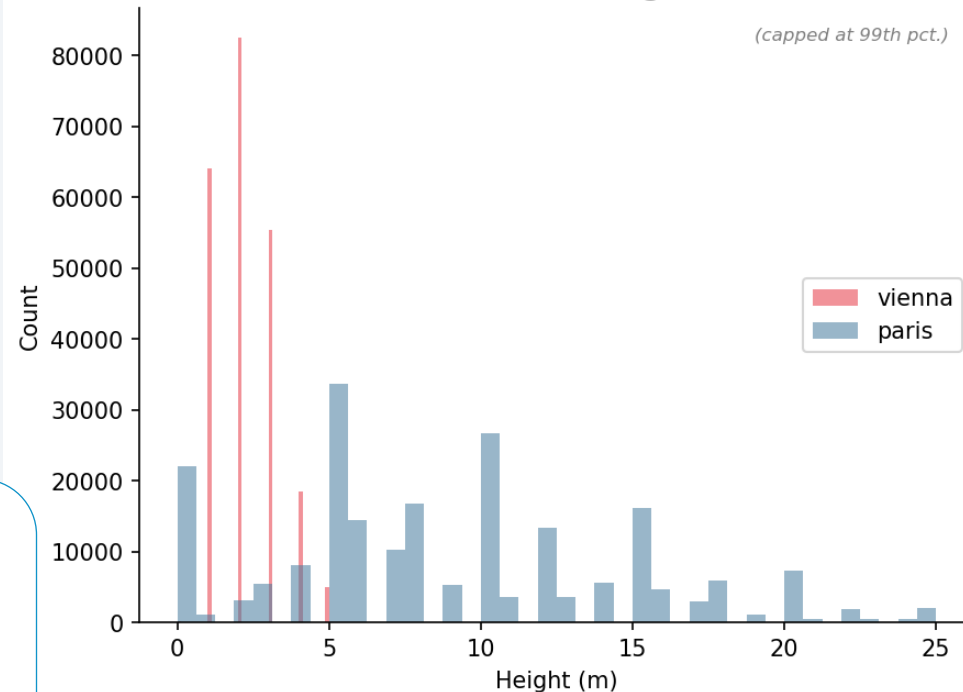
2. Vienna height & crown are encoded

- stored as ordinal classes 1–8 & 1–6, not continuous metres (mean 2.21 confirmed)
- Fix: decode to midpoint values (e.g. code 2 → 8.0 m), will be applied in data cleaning step before training

BAUMHOEHE	BAUMHOEHE_TXT	\
2	6–10 m	
4	16–20 m	
2	6–10 m	
2	6–10 m	
0	nicht bekannt	

Distribution of Height (m)

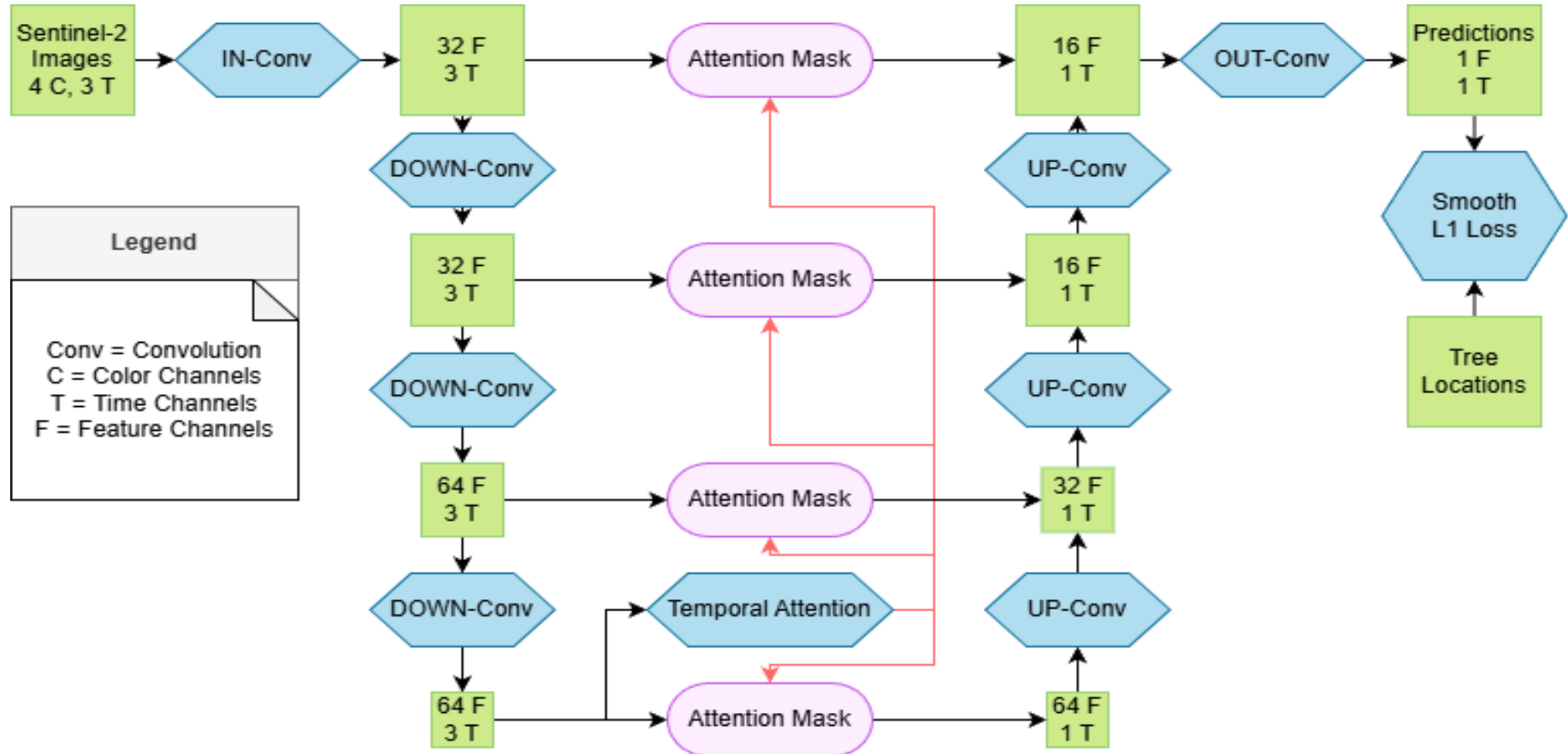
(capped at 99th pct.)



Machine Learning Pipeline



Model



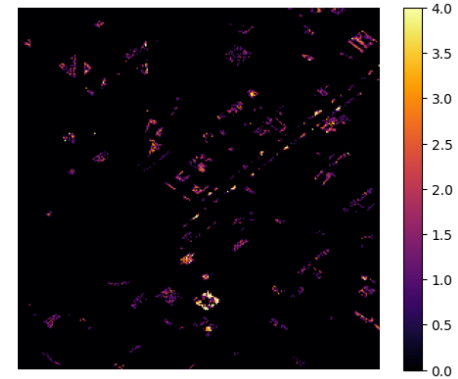
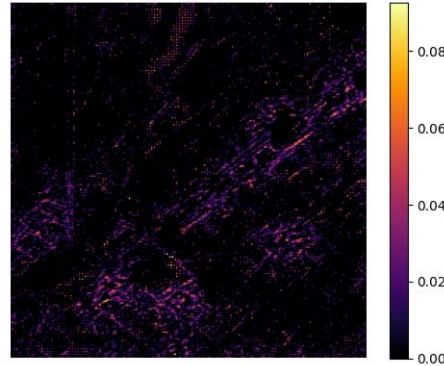
First results

Sentinel-2 Image

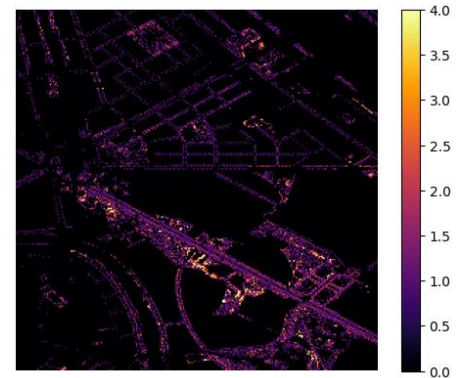
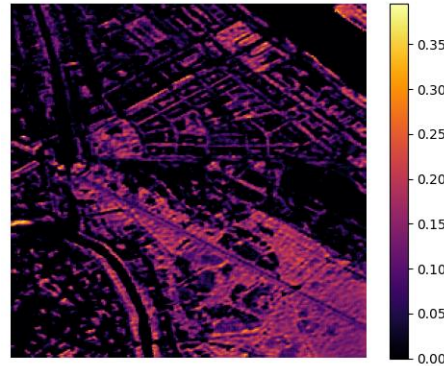
Model Prediction

Ground Truth

Barcelona



Vienna





Limited computational resources on WU Jupyter Hub



Sparse Coverage



Final database layout for Data Coaches

- Inclusion of OSM data
 - Classification: predicting different structure types
 - Regression: predicting number of trees using different structure types
- Train larger model for longer time
- Pick best model to create tree cadastre database
- Scrape new sentinel images to test model

Thanks for your attention

